

Data Structures

Course Information

Degree: UG

Program: BCA

Course Name: Data Structures

Course Credits: 4

Semester: II

Course Category: Core

Type of Course: Lecture

New Course: No

Learning Hours

Activity	Hours per Week
Lecture	2
Tutorial	0
Practical	2
Experiential Learning	3

Course Contribution

Category	Percentage Contribution
Employment Enhancement	45%
Entrepreneurship Skills	10%
Skill Development	45%

Course Relevance

Level	Percentage Relevance
Global	25%
National	25%
Regional	25%
Local	25%

Course Outcomes (COs)

CO#	Course Outcome	Bloom's Taxonomy Level			
CO1	Understand the Fundamentals of Data Structures.	BL-2			
CO2	Apply Searching and Sorting Techniques.	BL- 3			
CO3	Implement Stack and Queue Operations.	BL-2			
CO4	Implement Linked Lists.	BL-1			
CO5	Analyze and Implement Tree Data Structures.	BL-4			
Module	Course Content	Tools Used / Assessment and Activity	Bloom' s Level	Mapping COs	Percentage change in Module (put Nil if there is no change or it is a new course write NEW)
Module 1: Introduction on Data Structures	Introduction and Overview: Definition	Activity: Online Certification Coursera – 13 to 15 Hours	BL-1	1	NIL
	Classification and Operations of Data Structures.		BL-3	1	
	Recursion: Definition, Recursive Notation, Runtime Stack.		BL-3	1	
	Applications of Recursion: Factorial of Number, GCD, Fibonacci Series.		BL-4	1	
	Introduction to Dynamic Memory allocation (malloc(), calloc(), etc).		BL-3	1	
Module 2: Searching and Sorting	Binary Search.	Assessment: Internal Test Activity: Flip Class/ Mini project/ Presentation / Seminar	BL-2	2	NIL
	Sorting: Bubble Sort.		BL-3	2	
	Selection Sort.		BL-3	2	
	Insertion Sort.		BL-4	2	
	Merge Sort, Quick Sort		BL-4	2	

Module 3: Stacks and Queues	Stacks: Definition, Representation of Stacks using Arrays, Operations on Stacks using Arrays	Activity: Presentation / Case studies.	BL-3	3	NIL
	Application of Stacks: Arithmetic Expressions, Polish Notation, Conversion of Infix Expression to Postfix Expression.		BL-4	3	
	Queues: Definition, Representation of Queues using Array,		BL-2	3	
	Types of Queue: Simple Queue, Circular Queue		BL-4	3	
	Operations on Simple Queue using Array		BL-4	3	
Module 4: Linked List	Linked Lists: Definition	Activity: Presentation/Case studies	BL-2	4	NIL
	Comparison with Arrays, Representation		BL-4	4	
	Types of Linked lists		BL-3	4	
	Traversing, Inserting, and Deleting in Singly Linked List		BL-3	4	
Module 5: Trees	Trees: Definition	Assessment: Preparatory Exam	BL-1	5	NIL
	Terminology	Activity: Presentation /Case studies.	BL-1	5	
	Binary Trees		BL-2	5	
	Traversal of Binary Tree: Preorder, Inorder and postorder		BL-3	5	
	Binary Search Tree, Inserting, Deleting and Searching in Binary Search Tree		BL-4	5	

Textbooks:	
1	"Data Structures Using C" by Aaron M. Tenenbaum, Yedidyah Langsam, and Moshe J. Augenstein.
2	"Data Structures Through C" by Yashavant Kanetkar
3	"Data Structures and Algorithm Analysis in C" by Mark Allen Weiss
4	"C Programming with Data Structures" by Balagurusamy

CO-PO/PSO Mapping

CO	P01	P02	P03	P04	P05	P06	P07	P08	P09
C01	2	1			1		1		1
C02	3	2	2			2			
C03	3		2	2			1		
C04		2	3	2	3			1	
C05	2	3	3		2	2	2	1	2